

Anterior Triangle of the Neck

Anterior Triangle of the Neck: An anatomical region in the front of the neck defined by specific boundaries.

Boundaries:

Midline of the Neck: The central line that divides the neck into left and right halves.

Anterior Border of the Sternocleidomastoid Muscle: The front edge of a prominent muscle in the neck that helps in rotating and flexing the head.

Lower Border of the Mandible: The bottom edge of the jawbone that supports the lower teeth.

Submandibular Triangle / Digastric Triangle

Submandibular Triangle: An area located below the mandible, named for its contents and shape.

Digastric Triangle: An alternative name for the submandibular triangle, derived from its borders formed by the two bellies of the digastric muscle.

Boundaries:

Anterior Belly of the Digastric Muscle: The front portion of the digastric muscle, which assists in lowering the mandible.

Posterior Belly of the Digastric Muscle: The back portion of the digastric muscle.

Body of the Mandible: The main horizontal part of the jawbone.

Contents:

Submandibular Gland: A salivary gland located beneath the mandible that produces saliva.

Facial Artery and Vein: Blood vessels that supply and drain blood from the face.

Lymph Nodes: Small, bean-shaped structures that filter lymph and are part of the immune system.

Carotid Triangle

Carotid Triangle: A triangular area in the neck containing important blood vessels and nerves.

Boundaries:

Superior Belly of the Omohyoid Muscle: The upper part of a muscle that connects the shoulder to the hyoid bone.

Stylohyoid Muscle: A muscle that elevates the hyoid bone and assists in swallowing.

Contents:

Common Carotid Artery: A major blood vessel that supplies blood to the head and neck, which bifurcates (divides) into the external and internal carotid arteries.

Internal Jugular Vein: A large vein that drains blood from the brain and face.

Vagus Nerve (CN X): A cranial nerve that controls various autonomic functions, including heart rate and digestion.

Accessory Nerve (CN XI): A cranial nerve that controls certain muscles in the neck and shoulder.

Hypoglossal Nerve (CN XII): A cranial nerve responsible for tongue movement.

Clinical Relevance:

Vascular Surgeries: Surgical procedures involving blood vessels.

Palpation: The technique of examining the body through touch.

Carotid Artery Stenosis: A narrowing of the carotid artery that can lead to reduced blood flow to the brain.

Muscular Triangle

Muscular Triangle: An area of the neck that contains several important structures.

Boundaries:

Superior Belly of the Omohyoid Muscle: The upper portion of the omohyoid muscle.

Lower Border of the Sternocleidomastoid Muscle: The bottom edge of the sternocleidomastoid muscle.

Contents:

Thyroid Gland: A butterfly-shaped gland located in the neck that produces hormones regulating metabolism.

Parathyroid Glands: Small glands located near the thyroid that regulate calcium levels in the blood.

Trachea: The windpipe that carries air to and from the lungs.

Esophagus: The tube that carries food from the throat to the stomach.

Sublingual Triangle

Sublingual Triangle: An area located under the tongue.

Boundaries:

Body of the Mandible: The horizontal part of the lower jawbone.

Anterior Belly of the Digastric Muscle: The front portion of the digastric muscle.

Contents:

Sublingual Gland: A salivary gland located under the tongue.

Lingual Nerve: A branch of the mandibular nerve that provides sensation to the tongue.

Hypoglossal Nerve (CN XII): A cranial nerve that controls tongue movement.

Submandibular Triangle

Submandibular Triangle: An anatomical area located beneath the mandible (lower jaw).

Location

Situated Beneath the Mandible: Located under the lower jawbone.

Boundaries

Anterior Boundary:

Anterior Belly of the Digastric Muscle: The front portion of the digastric muscle, which assists in lowering the mandible.

Posterior Boundary:

Posterior Belly of the Digastric Muscle: The back portion of the digastric muscle.

Superior Boundary:

Lower Border of the Mandible: The bottom edge of the jawbone that supports the lower teeth.

Contents

Gland:

Submandibular Gland: A major salivary gland located within this triangle that produces saliva.

Nerves:

Hypoglossal Nerve (CN XII): A cranial nerve that innervates (supplies nerves to) the muscles of the tongue.

Lingual Nerve: A branch of the mandibular nerve that supplies sensory innervation to the floor of the mouth and the tongue.

Blood Vessels:

Facial Artery: A blood vessel that supplies blood to the face; it branches into smaller arteries, including the submental artery.

Facial Vein: A vein that drains blood from the face back to the heart.

Lymph Nodes:

Submandibular Lymph Nodes: Small structures that filter lymph fluid and drain lymph from the oral cavity, teeth, and surrounding areas.

Clinical Significance

Surgical Access:

Important in Procedures: Understanding this triangle is crucial for surgical procedures involving the submandibular gland and nearby structures.

Infections:

Awareness of This Triangle: Essential for diagnosing infections like sialadenitis (inflammation of the salivary gland) or abscesses (localized infections).

Relations

Superficial Structures:

Covered by the Skin and Platysma Muscle: The triangle is located beneath the skin and a thin muscle that covers the neck.

Deep Structures:

Related to the Mylohyoid Muscle: The mylohyoid muscle forms the floor of the mouth and is located deep to the submandibular triangle.

Occipital Triangle

Occipital Triangle: An anatomical region in the neck located near the back of the head.

Boundaries

Anterior Boundary:

Posterior Border of the Sternocleidomastoid Muscle: The back edge of a muscle that rotates and flexes the head.

Posterior Boundary:

Anterior Border of the Trapezius Muscle: The front edge of a large muscle that extends from the back of the neck to the shoulders and down the spine.

Inferior Boundary:

Superior Belly of the Omohyoid Muscle: The upper part of a muscle that helps depress the hyoid bone and is located in the neck.

Contents

Nerves:

Accessory Nerve (CN XI): A cranial nerve that innervates (supplies nerves to) the sternocleidomastoid and trapezius muscles, helping in head movement.

Cervical Plexus Branches:

Lesser Occipital Nerve: A nerve that provides sensory innervation to the skin behind the ear.

Great Auricular Nerve: A nerve that supplies sensation to the skin over the parotid gland (salivary gland near the ear).

Transverse Cervical Nerve: A nerve that provides sensory innervation to the anterior (front) part of the neck.

Vessels:

Occipital Artery: A branch of the external carotid artery that supplies blood to the posterior scalp (back of the head).

Transverse Cervical Artery: An artery that supplies blood to the muscles of the neck and shoulder.

External Jugular Vein: A vein that drains blood from the posterior part of the head and neck back to the heart.

Muscles:

Splenius Capitis: A muscle that extends (straightens) and rotates the head.

Levator Scapulae: A muscle that elevates (lifts) the scapula (shoulder blade) and assists in neck extension.

Scalene Muscles:

Anterior, Middle, and Posterior: A group of three muscles in the neck that aid in neck flexion (bending the neck forward) and assist with respiration (breathing) by elevating the ribs.

Clinical Significance

Nerve Blocks:

Useful for Performing Regional Anesthesia: Important for providing pain relief during neck surgeries.

Surgical Access:

Important Landmark in Surgeries: This triangle serves as a guide for surgeries involving the neck and posterior head.

Lymphatic Drainage:

Involves Lymph Nodes: These nodes are crucial for assessing infections or tumors in the head and neck region.

Relations

Superficial Structures:

Lies Deep to the Platysma Muscle: Located beneath a thin muscle that covers the neck and is also covered by skin.

Deep Structures:

Related to the Cervical Plexus: Involves nerve networks that supply the neck, brachial plexus roots (nerves that supply the upper limb), and other major blood vessels in the neck.

Subclavian Triangle

Subclavian Triangle: A small lower part of the posterior triangle of the neck located above the clavicle.

Boundaries of Posterior Triangle

Posterior Triangle: The area behind the sternocleidomastoid muscle.

Anterior:

Posterior Border of Sternocleidomastoid: The back edge of the muscle that helps turn and flex the head.

Posterior:

Anterior Border of Trapezius: The front edge of a large muscle extending from the back of the neck to the shoulder and spine.

Inferior (Base):

Middle One-Third of Clavicle: The central part of the collarbone.

Apex:

Superior Nuchal Line: A ridge on the occipital bone of the skull where the trapezius and sternocleidomastoid muscles meet.

Roof

Investing Layer of Deep Cervical Fascia: A layer of connective tissue that covers and supports structures in the neck.

Floor

Prevertebral Layer of Deep Cervical Fascia: A layer of fascia covering the muscles next to the vertebrae in the neck.

Covers Muscles:

Splenius Capitis: A muscle that extends and rotates the head.

Levator Scapulae: A muscle that elevates the scapula (shoulder blade) and assists in neck movement.

Scalenus Medius: A muscle involved in neck flexion and respiration.

Scalenus Posterior: A muscle that may be present and assists with neck flexion.

Contents of Subclavian Triangle

Nerves:

Three Trunks of Brachial Plexus: Major nerve branches that emerge between the anterior and middle scalene muscles, supplying the upper limb.

Nerve to Serratus Anterior (C5, C6, C7): A nerve that innervates the serratus anterior muscle, which helps in moving the shoulder.

Nerve to Subclavius (C5, C6): A nerve that innervates the subclavius muscle, located beneath the clavicle.

Accessory Phrenic Nerve: A nerve that may arise from the nerve to subclavius and joins the phrenic nerve, assisting in diaphragm movement for breathing.

Suprascapular Nerve (C5, C6): A nerve originating from the upper trunk of the brachial plexus that supplies the supraspinatus and infraspinatus muscles, located above the scapula.

Vessels:

Subclavian Artery: The major artery supplying blood to the upper limb, located behind the tendon of the scalenus anterior muscle.

Subclavian Vein: The vein that drains blood from the upper limb, located in front of the tendon.

Suprascapular Artery: A branch of the thyrocervical trunk that supplies blood to the shoulder area and passes behind the clavicle.

Transverse Cervical Artery: A branch of the thyrocervical trunk that crosses the scalenus anterior muscle, the phrenic nerve, and the upper trunks of the brachial plexus, dividing into superficial and deep branches.

Lymph Nodes

Supraclavicular Lymph Nodes: Lymph nodes located above the clavicle that are important for assessing infections or diseases in the head and neck region.

Carotid Sheath

Carotid Sheath: A connective tissue structure in the neck that encases important vascular and nervous components.

Formation

Deep Cervical Fascia: A layer of connective tissue in the neck that supports and separates structures.

Investing Layer of Cervical Fascia: The outermost layer of deep cervical fascia that surrounds various neck structures.

Attachments

Base of the Skull: The lower part of the skull that connects to the cervical spine.

Superior Mediastinum: The upper part of the mediastinum, located between the lungs and containing vital structures like the heart and great vessels.

Encases: Surrounds or envelops structures within it.

Relations

Anteriorly:

Common Carotid Artery: The major artery supplying blood to the head and neck, which bifurcates (divides) into the internal and external carotid arteries.

Internal Jugular Vein: The vein responsible for draining blood from the brain and face back to the heart.

Laterally:

Sternocleidomastoid Muscle: A prominent neck muscle that helps rotate and flex the head.

Medially:

Trachea: The windpipe that carries air to and from the lungs.

Esophagus: The tube that carries food from the throat to the stomach.

Posteriorly:

Prevertebral Fascia: A layer of connective tissue that covers the vertebrae and muscles in the back of the neck.

Cervical Spine: The portion of the spine that runs through the neck, made up of cervical vertebrae.

Content

Common Carotid Artery (and its bifurcation): The artery that divides into the internal and external carotid arteries, supplying blood to the head and neck.

Internal Jugular Vein: The vein that drains blood from the head and neck back to the heart.

Vagus Nerve (CN X): A cranial nerve that controls various autonomic functions, including heart rate and digestion.

Sympathetic Nerves: Nerves that originate from the sympathetic nervous system, which is responsible for the "fight or flight" response; they include cervical sympathetic ganglia.

Lymph Nodes (cervical lymph nodes): Clusters of lymphatic tissue in the neck that help filter lymph and play a role in the immune response.

Sternocleidomastoid

Sternocleidomastoid: A major muscle in the neck that plays a crucial role in head movement and neck flexion.

Attachment

Origin:

Sternal Head: The part of the muscle that starts from the front surface of the manubrium sterni, which is the uppermost part of the sternum (breastbone).

Clavicular Head: The portion that begins at the upper surface of the medial third of the clavicle (collarbone).

Insertion:

Lateral Surface of the Mastoid Process: The area on the skull behind the ear where the muscle attaches; the mastoid process is a bony prominence of the temporal bone.

Superior Nuchal Line: A ridge on the occipital bone at the back of the skull where the muscle also attaches.

Nerve Supply

Innervated: Supplied with nerve fibers that allow for muscle movement.

Accessory Nerve (CN XI): The cranial nerve responsible for motor control of the sternocleidomastoid and trapezius muscles.

Sensory Innervation: Refers to the sensation supplied to the area.

Cervical Plexus (C2 and C3): A network of nerves in the neck that provides sensation to the skin and muscles.

Actions

Bilateral Contraction: When both sides of the muscle contract.

Flexes the Neck Forward: Bends the neck forward at the cervical spine.

Extends the Head at the Atlanto-occipital Joint: Moves the head backward at the joint between the skull and the first cervical vertebra.

Unilateral Contraction: When one side of the muscle contracts.

Rotates the Head to the Opposite Side: Turns the head towards the side opposite the contracting muscle.

Tilts the Head to the Same Side: Bends the head towards the side of the contracting muscle.

Assists in Elevating the Thorax During Forced Inhalation: Helps lift the chest when breathing heavily.

Relations

Anteriorly:

Covers the Internal Jugular Vein: The large vein that drains blood from the head and neck.

Common Carotid Artery: The main artery supplying blood to the head and neck.

Posteriorly:

Related to the Trapezius Muscle: A large muscle in the upper back that helps move the shoulder and support the arm.

Medially:

In Relation to the Phrenic Nerve: The nerve that innervates the diaphragm, crucial for breathing.

Brachial Plexus: A network of nerves that innervates the arm and hand.

Laterally:

Lies Close to the External Jugular Vein: A vein that drains blood from the face and neck.

Inferiorly:

Overlies Structures of the Anterior Triangle of the Neck: The area containing vital structures like the thyroid gland and trachea.

External Jugular Vein

External Jugular Vein: A major vein that drains blood from the face and neck back to the heart.

Formation

Union: The joining together of two or more structures.

Posterior Auricular Vein: A vein that drains blood from the area behind the ear.

Retromandibular Vein: A vein formed by the confluence of the superficial and deep divisions of the mandibular vein, draining blood from the face and neck.

Emerges: To come into view or to become visible.

Posterior Aspect: The back side of a structure.

Angle of the Mandible: The point where the lower jaw (mandible) bends or turns upward.

Course

Passes: To move through a space.

Downward and Anteriorly: Moving towards the feet (downward) and towards the front of the body (anteriorly).

Across the Sternocleidomastoid Muscle: The external jugular vein crosses over the muscle that runs along the side of the neck.

Superficial: Located near the surface of the body.

Deep: Located further from the surface.

Platysma Muscle: A thin muscle that lies just beneath the skin of the neck and is involved in facial expressions.

Travels: Moves along a particular path.

Vertical Direction: Moving up and down in a straight line.

Tributaries: Smaller veins that drain into a larger vein, contributing blood flow.

Cervical Region: The area of the neck.

Facial Regions: Areas of the face.

Termination

Drains: To remove fluid or waste from an area.

Subclavian Vein: A large vein that carries blood from the arm and neck back to the heart.

Enters: To go into or become part of a structure.

Above the Clavicle: Located higher than the collarbone (clavicle).

Sometimes: Occasionally; not always.

Internal Jugular Vein: A major vein that drains blood from the brain and face back to the heart.